ORIGINAL ARTICLE

Daclizumab HYP versus Interferon Beta-1a in Relapsing Multiple Sclerosis

L. Kappos, H. Wiendl, K. Selmaj, D.L. Arnold, E. Havrdova, A. Boyko, M. Kaufman, J. Rose, S. Greenberg, M. Sweetser, K. Riester, G. O'Neill, and J. Elkins

The authors' full names, academic degrees, and affiliations are listed in the Appendix. Address reprint requests to Dr. Kappos at the Neurologic Clinic and Policlinic, the Departments of Medicine, Clinical Research, and Biomedicine and Biomedical Engineering, University Hospital, Basel, Petersgraben 4, 4031 Basel, Switzerland, or at kapposl-pa@usb.ch.

N Engl J Med 2015;373:1418-28.
DOI: 10.1056/NEJMoa1501481
Copyright © 2015 Massachusetts Medical Society.

ABSTRACT

BACKGROUND
Daclizumab high-yield process (HYP) is a humanized monoclonal antibody that binds to CD25 (alpha subunit of the interleukin-2 receptor) and modulates interleukin-2 signaling. Abnormalities in interleukin-2 signaling have been implicated in the pathogenesis of multiple sclerosis and other autoimmune disorders.

METHODS
We conducted a randomized, double-blind, active-controlled, phase 3 study involving 1841 patients with relapsing–remitting multiple sclerosis to compare daclizumab HYP, administered subcutaneously at a dose of 150 mg every 4 weeks, with interferon beta-1a, administered intramuscularly at a dose of 30 μg once weekly, for up to 144 weeks. The primary end point was the annualized relapse rate.

RESULTS
The annualized relapse rate was lower with daclizumab HYP than with interferon beta-1a (0.22 vs. 0.39; 45% lower rate with daclizumab HYP; P<0.001). The number of new or newly enlarged hyperintense lesions on T_2-weighted magnetic resonance imaging (MRI) over a period of 96 weeks was lower with daclizumab HYP than with interferon beta-1a (4.3 vs. 9.4; 54% lower number of lesions with daclizumab HYP; P<0.001). At week 144, the estimated incidence of disability progression confirmed at 12 weeks was 16% with daclizumab HYP and 20% with interferon beta-1a (P=0.16). Serious adverse events, excluding relapse of multiple sclerosis, were reported in 15% of the patients in the daclizumab HYP group and in 10% of those in the interferon beta-1a group. Infections were more common in the daclizumab HYP group than in the interferon beta-1a group (in 65% vs. 57% of the patients, including serious infection in 4% vs. 2%), as were cutaneous events such as rash or eczema (in 37% vs. 19%, including serious events in 2% vs. <1%) and elevations in liver aminotransferase levels that were more than 5 times the upper limit of the normal range (in 6% vs. 3%).

CONCLUSIONS
Among patients with relapsing–remitting multiple sclerosis, daclizumab HYP showed efficacy superior to that of interferon beta-1a with regard to the annualized relapse rate and lesions, as assessed by means of MRI, but was not associated with a significantly lower risk of disability progression confirmed at 12 weeks. The rates of infection, rash, and abnormalities on liver-function testing were higher with daclizumab HYP than with interferon beta-1a. (Funded by Biogen and AbbVie Biotherapeutics; DECIDE ClinicalTrials.gov number, NCT01064401.)

The New England Journal of Medicine
Downloaded from nejm.org at INSERM DISC DOC on January 4, 2016. For personal use only. No other uses without permission.
Copyright © 2015 Massachusetts Medical Society. All rights reserved.

Daclizumab is a humanized monoclonal antibody that binds to the alpha subunit (CD25) of the high-affinity interleukin-2 receptor.[1,2] Daclizumab treatment prevents signaling through the high-affinity interleukin-2 receptor and increases the availability of interleukin-2 to signal at its intermediate-affinity receptor.[1,2]

The use of daclizumab in patients with multiple sclerosis was based initially on the hypothesis that it directly antagonizes activated CD25+ effector T cells, which have long been implicated as key mediators of the pathogenic effects of multiple sclerosis.[1,3] Notably, effector T-cell numbers and recall responses appear to be largely unaffected by daclizumab in vivo.[3] Other clinically important immunologic effects, including an increase in CD56^{bright} natural killer cells and a decrease in lymphoid-tissue inducer cells, have been observed.[4,5]

Daclizumab high-yield process (HYP) was developed for long-term subcutaneous administration and has less antibody-dependent cytotoxicity than earlier forms of daclizumab.[3] A randomized, placebo-controlled study showed that daclizumab HYP was efficacious as monotherapy over a period of 1 year in patients with relapsing–remitting multiple sclerosis.[6] Here, we report the results of the DECIDE study, which compared subcutaneous daclizumab HYP, administered at a dose of 150 mg every 4 weeks, with intramuscular interferon beta-1a (Avonex, Biogen), an approved therapy for relapsing multiple sclerosis, over a period of 2 to 3 years of treatment.

METHODS

STUDY OVERSIGHT

An advisory committee (see the Supplementary Appendix, available with the full text of this article at NEJM.org) participated in the study design. The advisory committee and an independent data and safety monitoring board provided study oversight in collaboration with the sponsors. Data were collected by the investigators, were analyzed by the sponsors, and remained confidential during the study. All the authors were involved in each stage of the manuscript development, made the decision to submit the manuscript for publication, and take responsibility for the accuracy and completeness of the data and analyses. The first draft of the manuscript was written by the first and last authors, with medical writing assistance paid for by the sponsors. The sponsors reviewed and provided feedback on the manuscript to the authors, who had full editorial control of the manuscript. The study was conducted and reported with fidelity to the study protocol, which is available at NEJM.org.

PATIENTS

Key eligibility criteria included a confirmed diagnosis of relapsing–remitting multiple sclerosis (according to the 2005 McDonald criteria[7]), an age of 18 to 55 years, cranial magnetic resonance imaging (MRI) showing lesions that were consistent with a diagnosis of multiple sclerosis, and a score of 0 to 5.0 on the Expanded Disability Status Scale (EDSS; scores range from 0 to 10.0, with higher scores indicating worse disability[8]). Eligible patients had to have one of the following: two or more clinical relapses within the previous 3 years, with at least one clinical relapse occurring in the 12 months before randomization; or one or more clinical relapses and at least one new lesion on MRI that was not associated with the clinical relapse within the previous 2 years, with at least one of these events occurring in the 12 months before randomization. Key exclusion criteria are listed in the Supplementary Appendix.

All the patients provided written informed consent. Consent for continued participation was also obtained from patients who had a confirmed relapse or progression of disability. The study was approved by the relevant central and local ethics committees and was conducted in accordance with the International Conference on Harmonisation guidelines for Good Clinical Practice and the principles of the Declaration of Helsinki.

STUDY DESIGN AND TREATMENT

Patients were randomly assigned, in a 1:1 ratio, to receive either subcutaneous daclizumab HYP at a dose of 150 mg every 4 weeks and intramuscular placebo once weekly or intramuscular interferon beta-1a at a dose of 30 μg once weekly and subcutaneous placebo every 4 weeks for at least 96 weeks and no more than 144 weeks (Fig. S1 in the Supplementary Appendix). Patients continued treatment up to week 144 or until the last enrolled patient completed week 96, which was the planned end of the study.

Patients were instructed to take prophylactic

The New England Journal of Medicine
Downloaded from nejm.org at INSERM DISC DOC on January 4, 2016. For personal use only. No other uses without permission.
Copyright © 2015 Massachusetts Medical Society. All rights reserved.

treatment for influenza-like symptoms during the first 24 weeks of therapy in order to reduce any potential for unblinding (since interferon beta therapy may be associated with influenza-like side effects). Randomization was conducted with the use of a centralized interactive voice-response system and stratified according to study site and prior use of interferon beta with the use of permuted-block randomization. All the patients and study personnel, including the treating neurologists, were unaware of the treatment assignments. All the efficacy assessments were performed by trained, certified,[9] examining neurologists or technicians who were not involved in other aspects of care of the patients in the study.

STUDY PROCEDURES AND END POINTS

Study visits occurred every 4 weeks and included in-clinic administration of daclizumab HYP (or matching placebo), dispensing of interferon beta-1a (or matching placebo) for at-home administration, and clinical and safety assessments. The EDSS score was determined at screening, at baseline, every 12 weeks thereafter, and during unscheduled visits for suspected relapse. Standardized MRI scans were obtained at baseline and at weeks 24, 96, and 144 and were evaluated in a blinded manner at a central MRI reading center (see the Supplementary Appendix). The 29-item Multiple Sclerosis Impact Scale, version 1 (MSIS-29; scores range from 1 to 100, with higher scores indicating greater physical or psychological effect of multiple sclerosis from the patient's perspective[10]), was administered at baseline and every 24 weeks thereafter. Unscheduled visits were performed to evaluate suspected relapses within 72 hours after symptom onset.

The primary end point was the annualized relapse rate over a period of 144 weeks. Relapses were defined as new or recurrent neurologic symptoms that were not associated with fever or infection and that lasted at least 24 hours; the symptoms had to be accompanied by new objective neurologic findings on examination by the examining neurologist that were confirmed by the independent neurologic evaluation committee.

Secondary efficacy end points were the number of new or newly enlarged hyperintense lesions on T_2-weighted MRI scans of the brain over a period of 96 weeks; the proportion of patients with confirmed progression of disability over a period of 144 weeks, which was defined as an increase of at least 1.0 point from a baseline score of at least 1.0 or an increase of at least 1.5 points from a baseline score of 0 on the EDSS that was confirmed at 12 weeks; the proportion of patients who did not have a relapse over a period of 144 weeks; and the proportion of patients with an increase from baseline (indicating increasingly negative effects of multiple sclerosis) of at least 7.5 points on the MSIS-29 physical subscale at 96 weeks. Additional end points are listed in the Supplementary Appendix.

STATISTICAL ANALYSIS

Efficacy analyses were based on the intention-to-treat population (i.e., patients who underwent randomization and received ≥1 dose of study drug) unless otherwise specified. The number of new or newly enlarged hyperintense lesions on T_2-weighted images at week 96 was assessed in all the patients with nonmissing postbaseline scan data. All the statistical tests were two-sided, with a significance level of 0.05.

The primary end point (the annualized relapse rate over a period of 144 weeks) was tested with the use of a negative binomial regression model, with adjustment for baseline relapse rate, history of interferon beta use (yes vs. no), baseline EDSS score (≤2.5 points vs. >2.5 points), and baseline age (≤35 years vs. >35 years). This analysis included relapses and time during the study up to the earliest of the following: the start of alternative medication for multiple sclerosis, 180 days after the receipt of the last dose, or the end of the treatment period.

Secondary end points were rank-ordered (in the order listed above) and were tested with the use of a sequential closed-testing procedure. If a comparison was not significant (at the 0.05 significance level), all lower-ranked end points were not considered to be statistically significant within the closed-testing procedure.

A negative binomial regression model, with adjustments for baseline volume of hyperintense lesions on T_2-weighted images, history of interferon beta use, and age, was used to evaluate the number of new or newly enlarged hyperintense lesions on T_2-weighted images at week 96, as compared with baseline. Analysis was based on observed data prior to switching to alternative multiple sclerosis medication. A Cox proportional-hazards model was used for the analysis of the proportion of patients who were relapse free and

The New England Journal of Medicine
Downloaded from nejm.org at INSERM DISC DOC on January 4, 2016. For personal use only. No other uses without permission.
Copyright © 2015 Massachusetts Medical Society. All rights reserved.

for the time to disability progression. Models were adjusted for history of interferon beta use, age, baseline EDSS value, and baseline relapse rate (relapse analysis only). For the analysis of progression, patients who had tentative disability progression at the visit at the end of the treatment period (or at the last EDSS assessment before initiation of alternative multiple sclerosis medication) without a confirmatory assessment were censored at the last EDSS assessment.

A logistic-regression model, with adjustments for baseline MSIS-29 physical subscale score, baseline Beck Depression Inventory II score, history of interferon beta use, and age, was used to evaluate the proportion of patients with clinically meaningful worsening on the MSIS-29 physical subscale. Missing data were imputed with the use of the mean of the nonmissing items (if <10 of 20 items were missing), or a random-effects model was used to impute the missing MSIS-29 physical subscale score if the patient was missing 10 or more of the 20 items.

A detailed description of the statistical methods and sensitivity analyses is provided in the Supplementary Appendix. There were 20 tertiary end points described in the protocol and statistical analysis plan (see the complete list and results in Table S2 in the Supplementary Appendix); these analyses were not adjusted for multiple comparisons and are considered to be exploratory.

Safety analyses were summarized with the use of descriptive statistics for all the patients who received a dose of the study drug (safety population). Safety data that were collected up to 180 days after the last dose of daclizumab HYP were included. During the study, the protocol was amended to provide additional guidance on evaluating and managing cutaneous events and increased monitoring and management of liver-function abnormalities (see the Supplementary Appendix).

RESULTS

PATIENTS

A total of 1841 patients (919 patients in the daclizumab HYP group and 922 in the interferon beta-1a group) were enrolled at 244 study sites in 28 countries and underwent randomization between May 11, 2010, and April 16, 2012. All the patients who underwent randomization received a dose of the study drug and were included in the intention-to-treat population. The characteristics at baseline were well balanced between the treatment groups (Table 1, and Table S1 in the Supplementary Appendix). The median duration of treatment was 108.7 weeks in the daclizumab HYP group and 111.4 weeks in the interferon beta-1a group. The overall rates of study withdrawal and treatment discontinuation were similar in the two treatment groups (Fig. S2 in the Supplementary Appendix).

EFFICACY RESULTS

Primary End Point

The adjusted annualized relapse rate was lower in the daclizumab HYP group than in the interferon beta-1a group (0.22 vs. 0.39, representing a 45% lower rate with daclizumab HYP; P<0.001) (Table 2). The results of the prespecified and post hoc sensitivity analyses of the primary end point were consistent with those of the primary analysis (see the Supplementary Appendix).

Secondary End Points

The findings were also significant for the first-ordered secondary end point. The number of new or newly enlarged hyperintense lesions on T_2-weighted images at week 96 was 54% lower in the daclizumab HYP group than in the interferon beta-1a group (P<0.001) (Table 2). The results of additional MRI-related outcomes, including the number of gadolinium-enhancing lesions, the number of new hypointense lesions on T_1-weighted images, and the median annualized percentage of brain-volume loss, are shown in Table S2 in the Supplementary Appendix.

The results were not significant for the second-ordered secondary end point. At week 144, the estimated percentage of patients who had disability progression confirmed at 12 weeks as measured by the EDSS was 16% in the daclizumab HYP group and 20% in the interferon beta-1a group (P=0.16) (Fig. 1B and Table 2). Prespecified sensitivity analyses of confirmed disability progression showed that censoring the data after a tentative disability progression (i.e., after a patient had documented worsening in the EDSS score but before a confirmatory visit took place) was more common in the interferon beta-1a group than in the daclizumab HYP group, and the analysis of confirmed disability progression was sensitive to assumptions made about disability progression in these patients with

The New England Journal of Medicine
Downloaded from nejm.org at INSERM DISC DOC on January 4, 2016. For personal use only. No other uses without permission.
Copyright © 2015 Massachusetts Medical Society. All rights reserved.

Table 1. Characteristics of the Intention-to-Treat Population at Baseline.*

Characteristic	Interferon Beta-1a (N=922)	Daclizumab HYP (N=919)
Age — yr	36.2±9.3	36.4±9.4
Female sex — no. (%)	627 (68)	625 (68)
White race — no. (%)†	828 (90)	823 (90)
Previous therapy — no. (%)		
Disease-modifying therapy‡	376 (41)	380 (41)
Interferon beta	311 (34)	308 (34)
Time since diagnosis — yr	4.1±4.7	4.2±5.0
Time since first symptoms — yr	6.9±6.3	7.0±6.3
No. of relapses in previous 12 mo	1.6±0.8	1.5±0.7
EDSS score§		
Mean	2.5±1.3	2.5±1.2
Median	2.2	2.0
MSIS-29 physical subscale score¶	21.9±19.2	21.5±19.7
No. of hyperintense lesions on T_2-weighted MRI‖	51.8±37.4	49.2±35.5
Gadolinium-enhancing lesions		
Mean no. of lesions**	2.3±5.9	2.0±5.9
≥1 lesions — no./total no. (%)	414/909 (46)	398/900 (44)

* Plus–minus values are means ±SD. The characteristics at baseline were well balanced between the treatment groups (nominal P>0.05). The intention-to-treat population included all the patients who underwent randomization and received at least one dose of study drug. Daclizumab high-yield process (HYP) was administered subcutaneously at a dose of 150 mg every 4 weeks, and interferon beta-1a was administered intramuscularly at a dose of 30 μg once weekly.

† Race was determined by the investigator.

‡ Disease-modifying therapy included any prior disease-modifying or immunomodulatory therapy for multiple sclerosis, such as interferon beta-1a, interferon beta-1b, glatiramer acetate, natalizumab, mitoxantrone, azathioprine, fumaric acid, laquinimod, cyclophosphamide, mycophenolic acid, fingolimod, teriflunomide, methotrexate, alemtuzumab, cladribine, immune globulin, or temsirolimus.

§ Scores on the Expanded Disability Status Scale (EDSS) range from 0 to 10.0, with higher scores indicating worse disability.[8]

¶ Data on the 29-item Multiple Sclerosis Impact Scale (MSIS-29) were missing for eight patients in each treatment group. Scores range from 1 to 100, with higher scores indicating a greater physical or psychological effect of multiple sclerosis from the patient's perspective.[10]

‖ Data were missing for 14 patients in the interferon beta-1a group and for 19 in the daclizumab HYP group.

** Data were missing for 13 patients in the interferon beta-1a group and for 19 in the daclizumab HYP group.

censored data (Table S3 in the Supplementary Appendix).

On the basis of the prespecified hierarchical testing plan, the results of the analyses of the third and fourth prespecified secondary end points were not considered to be significant. The estimated percentage of patients who were free from relapse at week 144 was 67% in the daclizumab HYP group and 51% in the interferon beta-1a group (Fig. 1A and Table 2). Clinically meaningful worsening (defined as an increase of ≥7.5 points) in the patient-reported physical effect of multiple sclerosis, as assessed with the use of the MSIS-29 physical subscale, was observed in 19% of the patients in the daclizumab HYP group and 23% of those in the interferon beta-1a group (Table 2). Analyses of tertiary end points are shown in Table S2 and Fig. S3 in the Supplementary Appendix.

SAFETY

The overall incidence of adverse events was similar in the daclizumab HYP group and the interferon beta-1a group (91% in each group) (Table 3). Adverse events, excluding relapse of multiple sclerosis, were reported in 90% of the patients in the daclizumab HYP group and in 89% of those in the interferon beta-1a group. The most common adverse events (in ≥10% of patients in either group), excluding relapse of multiple sclerosis, are listed in Table 3. A total of 14% of the patients in the daclizumab HYP group and 9% of those in the interferon beta-1a group discontinued the study treatment owing to adverse events, excluding relapse of multiple sclerosis (Table 3, and Table S4 in the Supplementary Appendix).

Serious adverse events (excluding relapse of multiple sclerosis) were reported in 15% of the patients in the daclizumab HYP group and in 10% of those in the interferon beta-1a group (Table 3). The most common serious adverse events (in ≥3 patients in either group) are listed in Table S5 in the Supplementary Appendix. Five patients died during the study (Table 3), including one in the daclizumab HYP group and four in the interferon beta-1a group; none of the deaths were considered by the investigators, while they were unaware of the treatment assignment, to be related to the study treatment.

Infections were reported in 65% of the patients in the daclizumab HYP group and in 57% of those in the interferon beta-1a group. Five patients (1%) in the daclizumab HYP group and three (<1%) in the interferon beta-1a group discontinued treatment owing to infection. Common infections (in ≥10% of the patients in either group) were nasopharyngitis (in 25% of the patients in the daclizumab HYP group and 21% of

The New England Journal of Medicine
Downloaded from nejm.org at INSERM DISC DOC on January 4, 2016. For personal use only. No other uses without permission.
Copyright © 2015 Massachusetts Medical Society. All rights reserved.

Table 2. Clinical End Points According to Treatment Group.*

End Point	Interferon Beta-1a (N=922)	Daclizumab HYP (N=919)	P Value
Primary end point			
Adjusted annualized relapse rate			
Rate (95% CI)	0.39 (0.35–0.44)	0.22 (0.19–0.24)	<0.001
Percent reduction vs. interferon beta-1a (95% CI)	45 (36–53)		
Secondary end points†			
New or newly enlarged hyperintense lesions on T_2-weighted images over period of 96 wk‡			
Adjusted mean no. (95% CI)	9.4 (8.5–10.5)	4.3 (3.9–4.8)	<0.001
Percent reduction vs. interferon beta-1a (95% CI)	54 (47–61)		
Disability progression confirmed at 12 wk at wk 144			
Estimated percent of patients§	20	16	0.16
Hazard ratio, daclizumab HYP vs. interferon beta-1a (95% CI)	0.84 (0.66–1.07)		
Proportion of patients free from relapse at wk 144			
Estimated percent of patients§	51	67	—
Hazard ratio for relapse, daclizumab HYP vs. interferon beta-1a (95% CI)	0.59 (0.50–0.69)		
Clinically meaningful worsening on the MSIS-29 physical subscale score at wk 96¶			
Estimated percent of patients with worsening	23	19	—
Percent reduction in the odds of worsening vs. interferon beta-1a (95% CI)	24 (5–40)		

* The number of patients is the intention-to-treat population for each treatment group, excluding patients with missing data for baseline covariates unless otherwise noted. CI denotes confidence interval.

† Secondary end points were rank-ordered (in the order listed here) and were tested with the use of a sequential closed-testing procedure. If a comparison did not indicate significance (at the 0.05 significance level), all lower-ranked end points were not considered to be statistically significant within the closed-testing procedure. Thus, the end points for the percentage of patients who were relapse free at week 144 and for clinical worsening on the basis of MSIS-29 physical subscale score were not tested because the P value for the secondary end point of disability progression confirmed at 12 weeks was 0.16.

‡ The analysis was performed in the subgroup of the intention-to-treat population who had a postbaseline scan, which included 841 patients in the interferon beta-1a group and 864 in the daclizumab HYP group.

§ The percentage of patients was estimated by means of the Kaplan–Meier product-limit method.

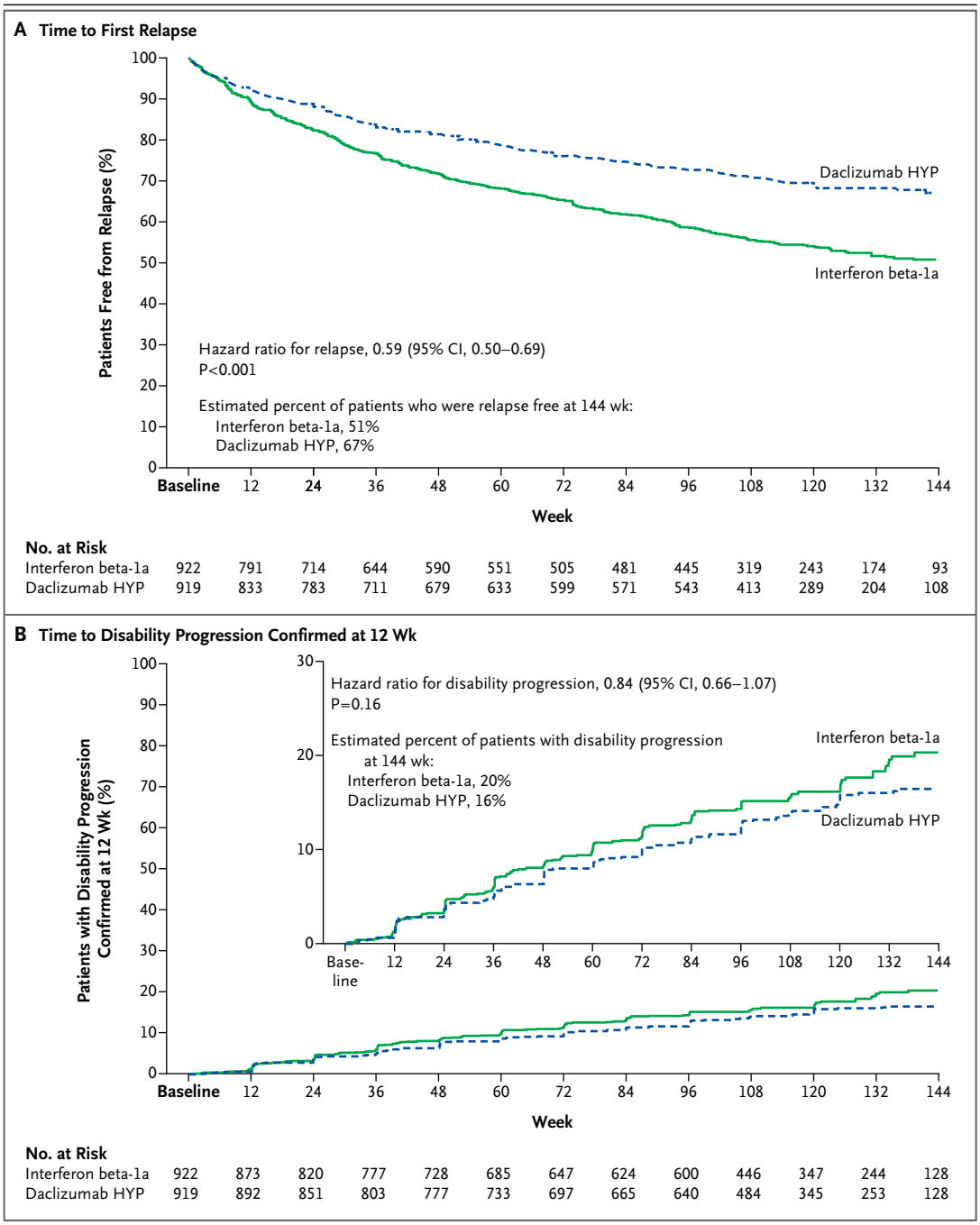
¶ Clinically meaningful worsening was defined as an increase of at least 7.5 points from baseline in the MSIS-29 physical subscale score. Data for 10 patients in the interferon beta-1a group and for 13 in the daclizumab HYP group were excluded owing to missing baseline covariates.

those in the interferon beta-1a group), upper respiratory tract infection (in 16% and 13%, respectively), and urinary tract infection (in 10% and 11%, respectively). The daclizumab HYP group and the interferon beta-1a group had a similar incidence of herpes virus infection (8% and 7%, respectively), including herpes zoster infection (1% and 1%, respectively).

Serious infection was reported in 4% of the patients in the daclizumab HYP group and in 2% of those in the interferon beta-1a group (Table S6 in the Supplementary Appendix). Serious infections that were reported in more than one patient during the study were urinary tract infection, pneumonia, appendicitis, cellulitis, and viral infection. No cases of progressive multifocal leukoencephalopathy or infectious encephalitis were reported during the study.

Cutaneous events (e.g., rash, eczema, seborrheic dermatitis, acne, erythema, and pruritus) were reported in 37% of the patients in the daclizumab HYP group and in 19% of those in the

The New England Journal of Medicine
Downloaded from nejm.org at INSERM DISC DOC on January 4, 2016. For personal use only. No other uses without permission.
Copyright © 2015 Massachusetts Medical Society. All rights reserved.



interferon beta-1a group and led to treatment discontinuation in 5% of patients and 1% of patients, respectively. The most common cutaneous events were rash (in 7% of patients in the daclizumab HYP group and 3% of those in the interferon beta-la group) and eczema (4% and 1%, respectively). Serious cutaneous events were reported in 2% of the patients in the daclizumab HYP group and in less than 1% of those in the interferon beta-1a group (Table S7 in the Supplementary Appendix). Serious cutaneous events that occurred in more than one patient were dermatitis and angioedema.

Hepatobiliary disorders were reported in 3% of the patients in the daclizumab HYP group and in 2% of those in the interferon beta-1a

The New England Journal of Medicine
Downloaded from nejm.org at INSERM DISC DOC on January 4, 2016. For personal use only. No other uses without permission.
Copyright © 2015 Massachusetts Medical Society. All rights reserved.

Figure 1 (facing page). Kaplan–Meier Plots of the Time to First Relapse and Time to Disability Progression Confirmed at 12 Weeks.

Panel A shows the estimated proportion of patients who were relapse free at week 144 on the basis of Kaplan–Meier estimates. Relapses were defined as new or recurrent neurologic symptoms that were not associated with fever or infection and that lasted at least 24 hours. The analysis included relapses that were confirmed by an independent neurology evaluation committee. Data were censored at the earliest of the following: start of alternative multiple sclerosis medication, 180 days after the last dose for patients who discontinued treatment, or the end of the treatment period for patients who completed treatment. Panel B shows the estimated proportion of patients with disability progression confirmed at 12 weeks in each treatment group on the basis of Kaplan–Meier estimates. Confirmed disability progression was defined as an increase of at least 1.0 point on the Expanded Disability Status Scale (EDSS; scores range from 0 to 10.0, with higher scores indicating worse disability) from a baseline score of at least 1.0 or as an increase of at least 1.5 points from a baseline score of 0; the increase had to be confirmed 12 weeks later. Data were censored at the time of withdrawal from the study or at the time a patient switched to alternative multiple sclerosis medication if the patient withdrew from the study or switched to alternative medication without disability progression. Patients who had tentative disability progression at the visit at the end of the treatment period (or at the last EDSS assessment before the start date of alternative multiple sclerosis medication) and who had no confirmation assessment were censored at the date of the last EDSS assessment. The inset shows the same data on an enlarged y axis. CI denotes confidence interval.

group, with serious hepatobiliary disorders occurring in 1% and less than 1%, respectively (Table 3). Hepatic events occurred in 16% of the patients in the daclizumab HYP group and in 14% of those in the interferon beta-1a group, with serious hepatic events occurring in 1% and less than 1%, respectively (Table 3, and Table S8 in the Supplementary Appendix).

Elevations in the alanine aminotransferase or aspartate aminotransferase level that were greater than 5 times the upper limit of the normal range occurred in 6% of the patients in the daclizumab HYP group and in 3% of those in the interferon beta-1a group (Table 3). Elevations of hepatic aminotransferase levels were observed more commonly during the first year of interferon beta-1a treatment than during the remainder of the study and were evenly distributed over time during daclizumab HYP treatment. Among patients who had elevations in the hepatic aminotransferase level of at least 3 times the upper limit of the normal range plus an elevation in the total bilirubin level greater than 2 times the upper limit of the normal range, one patient in each treatment group had a hepatic profile that was adjudicated by an independent committee as meeting the criteria of Hy's law[11] on the basis of a causality score of probable or higher[12] (Table 3).

No between-group differences were observed in the incidences of potentially clinically significant hematologic laboratory abnormalities (Table S9 in the Supplementary Appendix). During the study, malignant conditions were reported in seven patients in the daclizumab HYP group and in eight in the interferon beta-1a group (Table 3).

DISCUSSION

In this study involving a population of patients with active relapsing–remitting multiple sclerosis, 2 to 3 years of treatment with subcutaneous daclizumab HYP every 4 weeks was superior to the weekly administration of intramuscular interferon beta-1a with respect to relapse and lesion activity as assessed by means of MRI. Treatment with daclizumab HYP, as compared with interferon beta-1a, was not associated with a significantly lower risk of disability progression confirmed at 12 weeks.

Given the growing numbers of approved therapies, active-comparator studies are useful for evaluating new treatments for multiple sclerosis. When the treatment benefits of the active comparator are taken into account, the results of this study are consistent with the results of the 1-year, placebo-controlled SELECT study,[6] in which daclizumab HYP, as compared with placebo, was associated with a 54% lower relapse rate and a 57% lower risk of disability progression confirmed at 12 weeks.

Active-comparator studies are not without potential limitations. The risk of unblinding may be higher in these studies than in placebo-controlled studies, particularly when interferon beta is used as the comparator, given that it may be associated with influenza-like side effects. However, in this study, patients took prophylactic treatment for influenza-like symptoms, and the consistency of the clinical and MRI outcomes, as well as the similar treatment effect in

The New England Journal of Medicine
Downloaded from nejm.org at INSERM DISC DOC on January 4, 2016. For personal use only. No other uses without permission.
Copyright © 2015 Massachusetts Medical Society. All rights reserved.

patients with previous interferon exposure and those without previous exposure and in patients who had influenza-like symptoms and those who did not, suggest that unblinding was unlikely to have had a meaningful effect on the results.

In active-comparator studies, low event rates for end points such as confirmed disability progression can reduce statistical power. The analysis of disability progression can also be complicated by informative censoring that may occur when patients leave the study after having had a worsening in their EDSS score but before completing their confirmatory visit. Long-term follow-up studies will continue to be important in assessing the treatment effects of multiple

Table 3. Adverse Events.

Event	Interferon Beta-1a (N=922)	Daclizumab HYP (N=919)
	number (percent)	
Adverse event		
Any event	842 (91)	838 (91)
Any event, excluding multiple sclerosis relapse	816 (89)	823 (90)
Serious adverse event		
Any event	194 (21)	221 (24)
Any event, excluding multiple sclerosis relapse	88 (10)	142 (15)
Treatment discontinuation		
Due to adverse event	112 (12)	142 (15)
Due to adverse event, excluding multiple sclerosis relapse	84 (9)	131 (14)
Death*	4 (<1)	1 (<1)
Adverse events according to severity		
Mild	239 (26)	228 (25)
Moderate	495 (54)	483 (53)
Severe	108 (12)	127 (14)
Adverse events in ≥10% of either treatment group, excluding multiple sclerosis relapse		
Nasopharyngitis	197 (21)	226 (25)
Headache	175 (19)	159 (17)
Upper respiratory tract infection	124 (13)	149 (16)
Pyrexia	134 (15)	104 (11)
Injection-site pain	102 (11)	96 (10)
Urinary tract infection	98 (11)	96 (10)
Influenza-like illness	346 (38)	88 (10)
Adverse events of special interest		
Infection†	523 (57)	595 (65)
Serious infection	15 (2)	40 (4)
Cutaneous event†	176 (19)	344 (37)
Serious cutaneous event	1 (<1)	14 (2)
Hepatobiliary disorder†	16 (2)	26 (3)
Serious hepatobiliary disorder	4 (<1)	7 (1)
Hepatic event‡	130 (14)	144 (16)
Serious hepatic event‡	4 (<1)	6 (1)

The New England Journal of Medicine
Downloaded from nejm.org at INSERM DISC DOC on January 4, 2016. For personal use only. No other uses without permission.
Copyright © 2015 Massachusetts Medical Society. All rights reserved.

Table 3. (Continued.)

Event	Interferon Beta-1a (N=922)	Daclizumab HYP (N=919)
	number (percent)	
Hepatic laboratory abnormality§		
Alanine aminotransferase or aspartate aminotransferase ≥3× ULN	80 (9)	96 (10)
Alanine aminotransferase or aspartate aminotransferase >5× ULN	31 (3)	59 (6)
Alanine aminotransferase or aspartate aminotransferase ≥3× ULN and total bilirubin >2× ULN	1 (<1)	7 (1)
Met criteria of Hy's law	1 (<1)	1 (<1)
Malignant condition¶	8 (1)	7 (1)

* None of the deaths were considered by the investigators, while they were unaware of the treatment assignment, to be related to treatment. The death in the daclizumab HYP group occurred 117 days after withdrawal from treatment and was due to secondary complications of aspiration pneumonia in a patient who had a multiple sclerosis relapse involving the brain stem. Deaths in the interferon beta-1a group were due to myocardial infarction, suicide, metastatic pancreatic cancer, and peritonitis. In addition, one death due to multiple sclerosis progression was reported in each treatment group after study withdrawal. Neither of the deaths was considered to be related to study treatment.

† These events were assessed on the basis of the adverse events in the corresponding *Medical Dictionary for Regulatory Activities* System Organ Class.

‡ Hepatic events were defined with the use of the standardized query in the *Medical Dictionary for Regulatory Activities* for drug-related hepatic disorders that uses the comprehensive search, which includes a broad range of preferred terms, including hepatic laboratory abnormalities and hepatobiliary events.

§ Patients with an alanine or aspartate aminotransferase level that was at least 3 times the upper limit of the normal range (ULN) and a total bilirubin level that was greater than 2 times the ULN were reviewed and assessed by an independent panel with the use of a structured approach.[12] One patient in each group had a hepatic profile that was considered to meet the criteria of Hy's law[11] on the basis of a causality score of probable or higher. No patient had a causality score of definitive.

¶ Eight malignant conditions were reported in the interferon beta-1a group during the study: endometrial cancer, malignant melanoma, metastatic pancreatic carcinoma, squamous-cell carcinoma, cervical squamous-cell carcinoma, oral squamous-cell carcinoma, testicular seminoma, and a malignant tongue neoplasm (stage unspecified) in one patient each. Seven malignant conditions were reported in the daclizumab HYP group during the study: basal-cell carcinoma, malignant brain neoplasm, invasive ductal breast carcinoma, squamous-cell carcinoma of the lip, thyroid cancer, transitional-cell carcinoma, and uterine cancer in one patient each.

sclerosis therapies on disability progression, because studies of even 2 to 3 years' duration capture only a small fraction of the lifelong treatment that most patients with multiple sclerosis will require.

Interleukin-2 signaling plays a key role in both immunity against foreign antigens and maintenance of immune tolerance to self.[1] In this study, the risks of infection, cutaneous events, and elevations in hepatic aminotransferase levels were higher with daclizumab HYP than with interferon beta-1a. Why inflammatory events affecting organs such as the skin and liver also appeared to be facilitated by CD25 antagonism in some patients warrants further investigation. Although most of these events were self-limited, successful management of more serious cases will remain an important aspect of the overall risk–benefit profile of daclizumab HYP.

In conclusion, daclizumab HYP, a modulator of interleukin-2 signaling that was administered as a subcutaneous injection every 4 weeks, was associated with efficacy that was superior to that of weekly intramuscular injections of interferon beta-1a in the treatment of relapsing multiple sclerosis. The enhanced efficacy was accompanied by an increased frequency of adverse events, such that the net clinical benefit will need to be carefully considered by patients and their providers.

Supported by Biogen and AbbVie Biotherapeutics.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank the patients and all the DECIDE investigators for making this study possible; and Alison Gagnon and Elizabeth Cassell, Excel Scientific Solutions, for writing assistance and editorial assistance with earlier versions of the manuscript.

The New England Journal of Medicine
Downloaded from nejm.org at INSERM DISC DOC on January 4, 2016. For personal use only. No other uses without permission.
Copyright © 2015 Massachusetts Medical Society. All rights reserved.

APPENDIX

The authors' full names and academic degrees are as follows: Ludwig Kappos, M.D., Heinz Wiendl, M.D., Krzysztof Selmaj, M.D., Douglas L. Arnold, M.D., Eva Havrdova, M.D., Alexey Boyko, M.D., Michael Kaufman, M.D., John Rose, M.D., Steven Greenberg, M.D., Marianne Sweetser, M.D., Ph.D., Katherine Riester, M.P.H., Gilmore O'Neill, M.B., M.Med.Sc., and Jacob Elkins, M.D.

The authors' affiliations are as follows: the Neurologic Clinic and Policlinic, the Departments of Medicine, Clinical Research, and Biomedicine and Biomedical Engineering, University Hospital, Basel, Switzerland (L.K.); the Department of Neurology, University of Münster, Münster, Germany (H.W.); the Department of Neurology, Medical University of Lodz, Lodz, Poland (K.S.); NeuroRx Research and Montreal Neurological Institute, McGill University — both in Montreal (D.L.A.); the Department of Neurology, First Faculty of Medicine, Charles University in Prague, Prague, Czech Republic (E.H.); the Department of Neurology and Neurosurgery, Russian National Research Medical University, and Moscow Multiple Sclerosis Center — both in Moscow (A.B.); Cole Neurological Institute, University of Tennessee, Knoxville (M.K.); the Department of Neurology and the Neurovirology Research Laboratory, University of Utah, and the Veterans Affairs Salt Lake City Health Care System — both in Salt Lake City (J.R.); AbbVie Biotherapeutics, Redwood City, CA (S.G.); and Biogen, Cambridge, MA (M.S., K.R., G.O., J.E.).

REFERENCES

1. Wiendl H, Gross CC. Modulation of IL-2Rα with daclizumab for treatment of multiple sclerosis. Nat Rev Neurol 2013;9:394-404.

2. Martin R. Anti-CD25 (daclizumab) monoclonal antibody therapy in relapsing-remitting multiple sclerosis. Clin Immunol 2012;142:9-14.

3. Bielekova B. Daclizumab therapy for multiple sclerosis. Neurotherapeutics 2013;10:55-67.

4. Bielekova B, Catalfamo M, Reichert-Scrivner S, et al. Regulatory CD56(bright) natural killer cells mediate immunomodulatory effects of IL-2Ralpha-targeted therapy (daclizumab) in multiple sclerosis. Proc Natl Acad Sci U S A 2006;103:5941-6.

5. Perry JS, Han S, Xu Q, et al. Inhibition of LTi cell development by CD25 blockade is associated with decreased intrathecal inflammation in multiple sclerosis. Sci Transl Med 2012;4:145ra06.

6. Gold R, Giovannoni G, Selmaj K, et al. Daclizumab high-yield process in relapsing-remitting multiple sclerosis (SELECT): a randomised, double-blind, placebo-controlled trial. Lancet 2013;381:2167-75.

7. Polman CH, Reingold SC, Edan G, et al. Diagnostic criteria for multiple sclerosis: 2005 revisions to the "McDonald Criteria." Ann Neurol 2005;58:840-6.

8. Kurtzke JF. Rating neurologic impairment in multiple sclerosis: an Expanded Disability Status Scale (EDSS). Neurology 1983;33:1444-52.

9. Neurostatus Systems. Neurostatus training. 2007 (http://www.neurostatus.net).

10. Hobart J, Lamping D, Fitzpatrick R, Riazi A, Thompson A. The Multiple Sclerosis Impact Scale (MSIS-29): a new patient-based outcome measure. Brain 2001;124:962-73.

11. Temple R. Hy's law: predicting serious hepatotoxicity. Pharmacoepidemiol Drug Saf 2006;15:241-3.

12. Rockey DC, Seeff LB, Rochon J, et al. Causality assessment in drug-induced liver injury using a structured expert opinion process: comparison to the Roussel-Uclaf causality assessment method. Hepatology 2010;51:2117-26.

Copyright © 2015 Massachusetts Medical Society.

POSTING PRESENTATIONS FROM MEDICAL MEETINGS ONLINE

Online posting of an audio or video recording of an oral presentation at a medical meeting, with selected slides from the presentation, is not considered prior publication. Authors should feel free to call or send e-mail to the *Journal*'s Editorial Offices if there are any questions about this policy.

The New England Journal of Medicine
Downloaded from nejm.org at INSERM DISC DOC on January 4, 2016. For personal use only. No other uses without permission.
Copyright © 2015 Massachusetts Medical Society. All rights reserved.